

Financial Data EDA Report

Automated Financial MLOps Platform

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Analysis Type: Exploratory Data Analysis (EDA)

1. Introduction

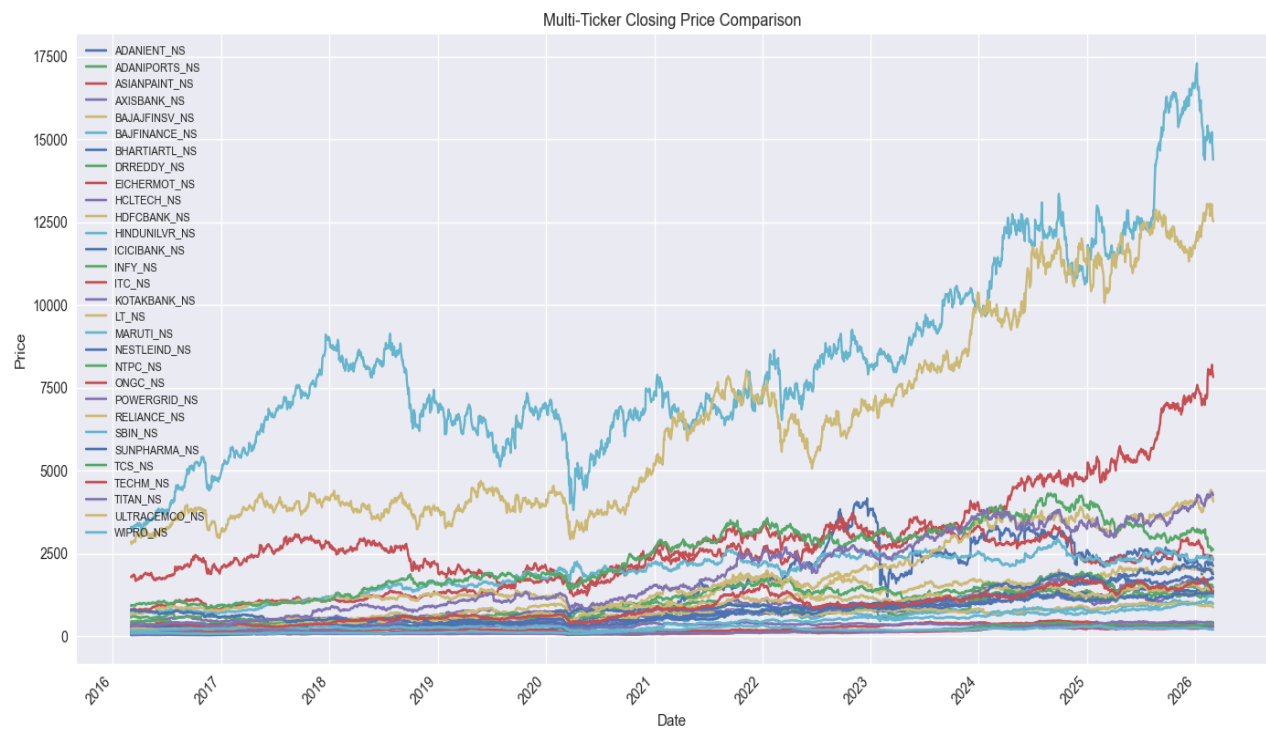
This report presents a structured exploratory analysis of financial datasets. The objective is to understand stock behavior and support machine learning models for predicting next-day price movement (UP/DOWN).

2. Top 10 Most Volatile Stocks

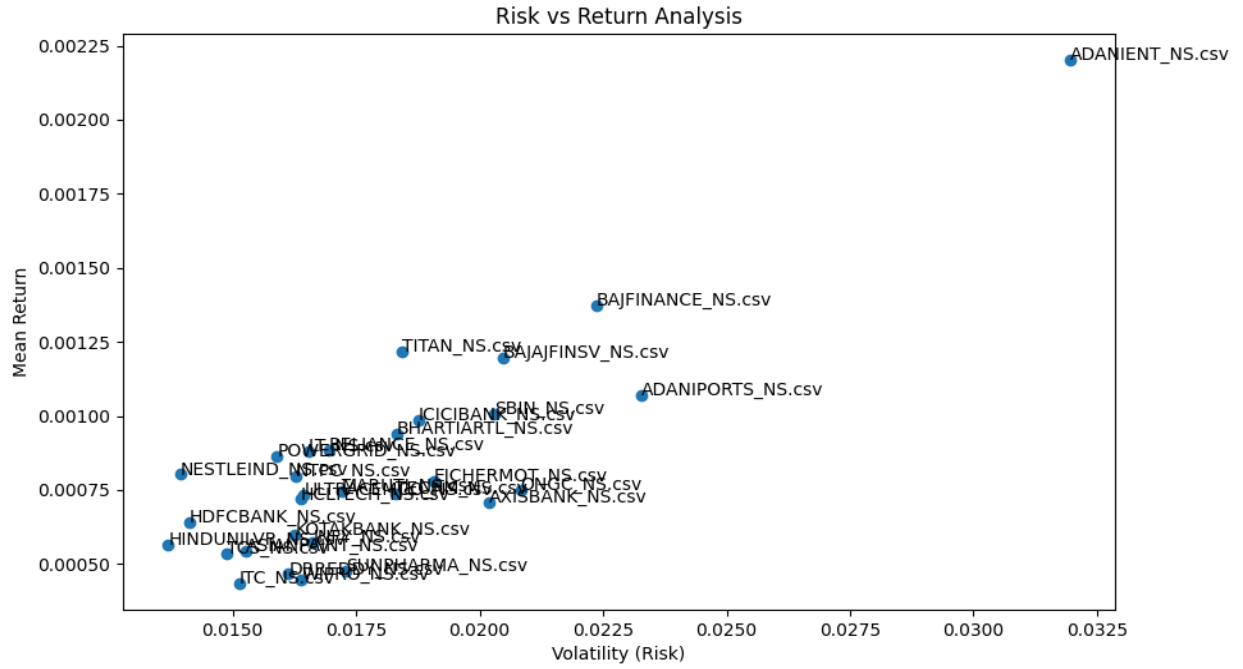
Dataset	Rows	Columns	Mean_Close	Std_Close	UP_Ratio	Volatility
ADANIENT_NS.csv	2470	7	1269.4204	1226.8362	0.5344	0.032
ADANIPORTS_NS.csv	2470	7	673.9036	405.7649	0.5134	0.0233
BAJFINANCE_NS.csv	2470	7	483.0262	276.9532	0.5206	0.0224
ONGC_NS.csv	2470	7	132.9429	62.5371	0.5121	0.0208
BAJAJFINSV_NS.csv	2470	7	1099.7773	585.6768	0.5166	0.0205
SBIN_NS.csv	2470	7	438.3687	236.4876	0.5219	0.0203
AXISBANK_NS.csv	2470	7	765.9328	256.5916	0.5105	0.0202
EICHERMOT_NS.csv	2470	7	3088.4225	1393.8852	0.5024	0.0191
ICICIBANK_NS.csv	2470	7	664.4927	390.6092	0.5065	0.0188
TITAN_NS.csv	2470	7	1875.1059	1175.3518	0.5223	0.0184

3. Market-Level Insights

Multi-Ticker Price Trend



Risk vs Return Analysis

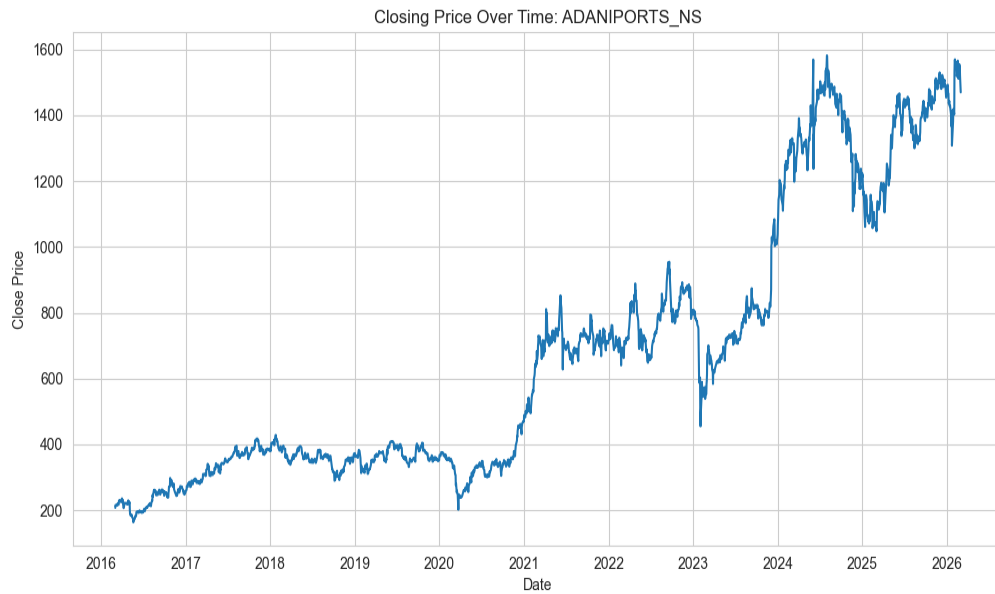


4. Sample Stock Price Trends

ADANIENT_NS



ADANIPORTS_NS



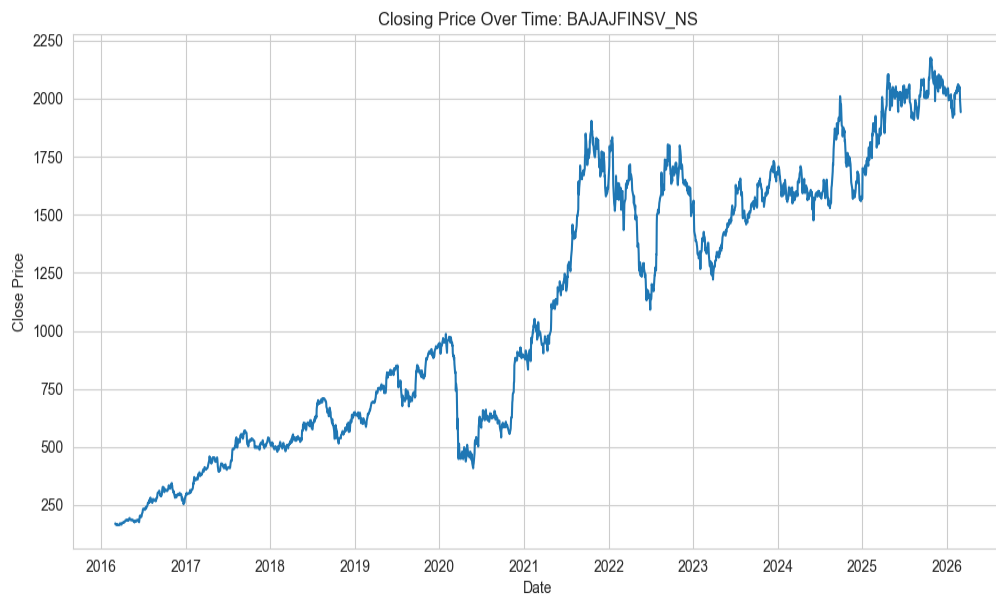
ASIANPAINT_NS



AXISBANK_NS



BAJAJFINSV_NS



5. Key Insights & Interpretation

- High volatility stocks indicate higher prediction uncertainty.
- UP ratio near 0.5 suggests near-random movement patterns.
- Low volatility stocks are more stable and easier to model.
- Correlation across stocks indicates sector-level dependencies.
- Risk vs Return highlights trade-offs critical for financial decision-making.

These insights guide feature engineering, model selection, and evaluation in the MLOps pipeline.